

Nathan Mwasuku

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Data Engineer with hands-on experience delivering scalable, production-grade data pipelines for analytics and machine learning use cases. Strong background in Python, SQL, and cloud-native architectures, with proven experience improving data quality, performance, and reliability across large-scale datasets. Experienced working in cross-functional, client-facing environments, translating business requirements into robust data solutions.

Skills

Data Engineering: Python, SQL, dbt

Cloud & Platforms: AWS, Snowflake, Databricks

Software & DevOps Practices: Git, CI/CD, Apache Airflow

Machine Learning Exposure: TensorFlow, Keras

Experience

Data Engineer | Digital Futures (Aviva)

Jan 2025 - Present

- Contributed to an enterprise-scale migration from Oracle to Snowflake, re-engineering 10,000+ lines of legacy SQL using advanced joins, aggregations, and window functions reducing execution times by **up to 40%**.
- Designed and implemented a Python-based data validation framework, ensuring **99.9%** data accuracy across Oracle-to-Snowflake reconciliation workflows.
- Collaborated closely with data analysts and business stakeholders to translate requirements into scalable, high-quality data models.
- Built and maintained **50+ dbt models** using advanced SQL and Jinja templating, delivering **millions of rows of business-critical data** daily to downstream stakeholders.
- Orchestrated production workflows using **Apache Airflow**, enabling reliable, automated scheduling of complex data transformations.
- Produced clear technical documentation to support knowledge sharing and long-term maintainability across engineering teams.

Sales Professional | SCS

Sept 2023 – July 2024

- Conducted in-depth client consultations to understand customer needs and translate requirements into tailored product recommendations, regularly exceeding monthly sales targets by ~15%.
- Communicated complex product features clearly and persuasively to non-technical audiences, strengthening client trust and satisfaction.
- Collaborated with design and logistics teams to deliver end-to-end customer solutions, balancing client expectations with operational constraints.
- Developed strong stakeholder management, negotiation, and time-management skills in a fast-paced, target-driven environment.

Projects

Pinterest Cloud-Based Data Pipeline Github Repo (click me)

- Designed and implemented a cloud-native data pipeline supporting both batch and real-time ingestion using **AWS Lambda, Kinesis, and MSK (Kafka)**.
- Built real-time streaming pipelines handling **sub-second latency** ingestion for high-volume event data.
- Developed REST APIs using **AWS API Gateway** to expose processed data to downstream services.
- **Engineered an AWS S3 data lake** with automated ingestion and scalable storage patterns.
- Processed and transformed large-scale datasets using **PySpark on Databricks**, enabling distributed analytics over millions of records.

ML-Driven Stock Price Forecasting | Dissertation Project - First Class

- Designed and trained **Artificial Neural Networks (ANNs)** using TensorFlow and Keras to forecast short-term stock price movements, achieving **~80% predictive accuracy**.
- Built robust data pipelines using **Pandas and NumPy** to clean and prepare large-scale financial datasets (100,000+ records).
- Applied **Object-Oriented design principles** to design a modular, reusable machine learning codebase, improving maintainability and reproducibility of experiments.

Education

Bachelor of Science in Mathematics Manchester Metropolitan University

Sept 2020 - June 2023

- First Class Honours

Certifications

AWS Certified Solutions Architect – Associate [click me](#)